

Name of the Student Abhishek KumarScholar Number 56

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY BHOPAL
DEPARTMENT OF ELECTRICAL ENGINEERING
MID TERM EXAMINATION-SEPTEMBER 2023

Course: B.Tech**Semester: III****Subject Name: Measurement and Instrumentation****Subject Code: EE-214****Time Duration: 1 Hour 30 Min****Maximum Marks: 30****Subject Coordinator Name: Dr. Sushma Gupta & Dr. More Raju**

S.No	Question	Marks	CO
1	A resistive temperature sensor with a nominal resistance (R_0) of 100Ω at 25 degrees Celsius. The temperature coefficient of the sensor material is 0.5% per degree Celsius. Calculate the resistance of the sensor at a temperature of 100 degrees Celsius?	5	CO-3
2	Describe the construction, operating principle and characteristic of linear variable displacement transducer.	10	CO-3
3	Classify the electrical type of instruments. Explain each one with suitable examples	5	CO-1
4	If V_{RN} , V_{YN} and V_{BN} are the instantaneous line to neutral voltages and i_R , i_Y and i_B are instantaneous line currents in a balanced three phase circuit, then compute $V_{RN}(i_Y - i_B) - (V_{YN} - V_{BN})i_R$ with a suitable phasor diagram.	5	CO-2
5	A periodic waveform observed across a load is represented by $V(t) = \begin{cases} 1 + \sin \omega t & 0 \leq \omega t < 6\pi \\ -1 + \sin \omega t & 6\pi \leq \omega t < 12\pi \end{cases}$ Find the measured value of voltage, if a moving iron voltmeter is connected across the load.	5	CO-2